

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q17-0838

Dresden Generating Station 500 kV
July 04, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0838. The Interconnection Customer proposes a 1214.0 MW natural gas combined cycle facility interconnecting at the Dresden Generating Station 500 kV substation in the Illinois area. The request was received on July 29, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 389 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$93,321,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
STA S VAN NUYS-MINNEHAHA 230 KV	TROY ENERGY LLC-RICHLAND 500 KV	125.7	786	17.1
ROBISON PARK SUBSTATION-DEKALB 230 KV	JONES CHAPEL-DEKALB 500 KV	113.9	1,515	34.5
MUSKOGEE-DARLINGTON RD 100 KV	RICHLAND-DEKALB SUBSTATION 500 KV	113.7	296	6.7
ADAMS-NEW DEPARTURE 230 KV	TREGO-BEAVER 500 KV	113.7	864	7.9
MASON ST 500-230 kV Transformer	TREGO-BEAVER 500 KV line	113.3	1,308	12.0
EUFULA-BEAVER 500 KV L	POST ROCK-NEWPORT TIE SUBSTATION 230 KV	112.8	3,064	28.5
MYRTLE PT-GALION GENERATING STA 230 KV	TREGO-BEAVER 500 KV	111.5	1,011	7.5
HIGHGATE-HERALD-TWIN CITIES 230 KV	MASON ST 500/230kV XFMR	111.5	857	7.0
EASTOWN RD-SWAN VLY SUBSTATION 230 KV	TREGO-SWAN VALLEY SUBSTATION 500 KV	111.0	614	6.3
GLION GENERATING STA 500/230 kV Auto	TREGO-BEAVER 500 KV	110.4	1,059	10.9
SIDNEY A MURRAY JR HYDROELECTRIC-24TH & GARDENDALE 230 KV	RICHLAND-DEKALB 500 KV	108.7	492	9.3
24TH & GARDENDALE-LANDSMAN CREEK 230 KV Line	DEKALB-RICHLAND 500 KV Line	107.9	475	9.3

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
WAYNE TRACE SUBSTATION-DARLINGTON RD 100 KV	RICHLAND-DEKLB 500 KV	107.7	339	6.0
SLOCUM RD-MCNEIL 230 KV	DEKALB-RICHLAND 500 KV L	106.5	1,072	25.9
SLOCUM RD-DEKALB 230 KV	RICHLAND-DEKALB 500 KV	106.2	1,101	25.9
MONTOUR STA-TALENENERGY MARTINS CRK LLC WILLIAMSPORT 345 KV	CORONET-FORT HUMBUG 500 KV	106.0	1,762	5.0
SWAN VLY-TREGO 500 KV	TROY ENERGY LLC-RICHLAND 500 KV	105.7	3,663	19.7
MUSKOGEE-MONTPELIER ELECTRIC GENERATING STA 500 KV L	LONETREE-SABANERA 500 KV	105.7	2,801	10.3
MINNEHAHA-NAPOLEON PEAKING STATION SUBSTATION 230 KV	TROY ENERGY LLC-RICHLAND 500 KV Line	105.2	794	13.2
TROY ENERGY LLC-RICHLAND 500 KV	SWAN VALLEY-TREGO 500 KV L	105.2	2,642	37.9
GRUVER-GALEVILLE 230 KV	IMPERIAL-EFAULA 500 KV	105.2	514	6.6
MCNEIL-BREED 230 KV	DEKALB-RICHLAND 500 KV line	104.8	1,103	25.9
SIDNEY A MURRAY JR HYDROELECTRIC-BREED 230 KV	RICHLAND-DEKALB 500 KV	104.8	1,103	25.9
IMPERIAL-EUAULA 500 KV	NEWPORT TIE SUBSTATION-POST OCK 230 KV	104.7	2,771	28.0
HERALD-TWIN CITIES-MASON ST 230 KV Line	NEW DEPARTURE-MASON ST 230 KV	104.6	610	5.5

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
STA S VAN NUYS-MINNEHAHA 230 KV	Rebuild line with higher-capacity conductor	\$9,931,000	14% (\$1,357,000)
ROBISON PARK SUBSTATION-DEKALB 230 KV	Rebuild line with higher-capacity conductor	\$31,095,000	14% (\$4,239,000)
MUSKOGEE-DARLINGTON RD 100 KV	Rebuild line with higher-capacity conductor	\$14,521,000	14% (\$1,980,000)
ADAMS-NEW DEPARTURE 230 KV	Rebuild line with higher-capacity conductor	\$28,504,000	14% (\$3,882,000)
MASON ST 500-230 kv Transformer	Replace transformer with higher-rated unit	\$31,074,000	14% (\$4,236,000)
EUFAULA-BEAVER 500 KV	Rebuild line with higher-capacity conductor	\$68,335,000	14% (\$9,296,000)
MYRTLE PT-GALION GENERATING STA 230 KV	Rebuild line with higher-capacity conductor	\$42,174,000	14% (\$5,741,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
HIGHGATE-HERALD-TWIN CITIES 230 KV	Rebuild line with higher-capacity conductor	\$30,531,000	14% (\$4,157,000)
SWAN VALLEY-EASTOWN ROAD 230 KV	Rebuild line with higher-capacity conductor	\$6,070,000	14% (\$828,000)
GALION GENERAING STATION 500-230 kv XFMR	Replace transformer with higher-rated unit	\$29,035,000	14% (\$3,955,000)
SIDNEY A MURRAY JR HYDROELECTRIC-24TH & GARDENDALE 230 KV	Reconductor line	\$12,414,000	14% (\$1,696,000)
24TH & GARDENDALE-LANDSMAN CRK 230 KV	Reconductor line	\$15,305,000	14% (\$2,091,000)
WAYNE TRACE-DARLINGTON RD 100 KV L	Reconductor line	\$551,000	14% (\$75,000)
SLOCUM RD-MCNEIL 230 KV Line	Reconductor line	\$5,754,000	14% (\$786,000)
SLOCUM RD-DEKALB SUBSTATION 230 KV	Reconductor line	\$15,123,000	14% (\$2,066,000)
MONTOUR STA-TALENENERGY MARTINS CRK LLC WILLIAMSPORT 345 KV	Reconductor line	\$30,442,000	26% (\$7,850,000)
SWAN VLY-TREGO 500 KV	Reconductor line	\$35,434,000	14% (\$4,812,000)
MUSKOGEE-MONTPELIER ELECTRIC GENERATING STA 500 KV L	Reconductor line	\$61,466,000	14% (\$8,340,000)
MINNEHAHA-NAPOLEON PEAKING STATION SUBSTATION 230 KV	Reconductor line	\$15,646,000	14% (\$2,140,000)
RICHLAND-TROY ENERGY LLC 500 KV	Reconductor line	\$98,285,000	14% (\$13,418,000)
GRUVER-GALEVILLE 230 KV	Reconductor line	\$4,372,000	14% (\$595,000)
BREED SUBSTATION-MCNEIL 230 KV	Reconductor line	\$5,529,000	14% (\$755,000)
SIDNEY A MURRAY JR HYDROELECTRIC-BREED 230 KV line	Reconductor line	\$19,071,000	14% (\$2,606,000)
IMPERIAL-EUAULA 500 KV	Reconductor line	\$39,096,000	14% (\$5,320,000)
HERALD-TWIN CITIES-MASON ST 230 KV Line	Reconductor line	\$8,055,000	14% (\$1,100,000)
	Total allocated to project		\$93,321,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.